Angio OMEX Combination Report — lchem=100

angio_lchem_sweep_combo001 | 10 replicate(s) | sim_time 100.0 MCS

Combination Overview

Property	Value
Scan point	lchem=100
Replicates	10 successful (plotted and averaged)
Signals	num_clusters, cell_density, largest_cluster_size, vegf_mean
Time	MCS (angio has no physical-time calibration)

Provenance (OMEX Metadata)

Field	Value
Creator (experimenter)	Joel Vanin
Created	2026-10-21T11:48:11
Modified	2026-10-21T11:48:21.035666
License	https://spdx.org/licenses/CC-BY-4.0.html
Citation	Merks, R.M.H., Brodsky, S.V., Goligorksy, M.S., Newman, S.A., Glazier, J.A. (2006). Cell elongation is key to in silico replication of in vitro vasculogenesis and subsequent remodeling. Developmental Biology 289(1), 44-54. https://doi.org/10.1016/j.ydbio.2005.10.003
Keywords	angiogenesis, vasculogenesis, cellular Potts model, VEGF, agent-based model

Run Identity (stochastic realization & code)

Replicate	RNG seed	Origin	Schema fingerprint	Code
angio_lchem_sweep_combo001_rep01	1186538411	drawn	sha256:0a6c2ceaca8b9cdf	built 493b273
angio_lchem_sweep_combo001_rep02	1591706957	drawn	sha256:0a6c2ceaca8b9cdf	built 493b273
angio_lchem_sweep_combo001_rep03	1517375581	drawn	sha256:0a6c2ceaca8b9cdf	built 493b273
angio_lchem_sweep_combo001_rep04	917905212	drawn	sha256:0a6c2ceaca8b9cdf	built 493b273
angio_lchem_sweep_combo001_rep05	1497126697	drawn	sha256:0a6c2ceaca8b9cdf	built 493b273
angio_lchem_sweep_combo001_rep06	1769884084	drawn	sha256:0a6c2ceaca8b9cdf	built 493b273
angio_lchem_sweep_combo001_rep07	515536928	drawn	sha256:0a6c2ceaca8b9cdf	built 493b273
angio_lchem_sweep_combo001_rep08	63575998	drawn	sha256:0a6c2ceaca8b9cdf	built 493b273
angio_lchem_sweep_combo001_rep09	579214122	drawn	sha256:0a6c2ceaca8b9cdf	built 493b273
angio_lchem_sweep_combo001_rep10	2061748046	drawn	sha256:0a6c2ceaca8b9cdf	built 493b273

Replicates within a combination share a parameterization and differ only in the random number generator seed, so a difference between them is stochastic rather than configured. Seeds marked *drawn* were generated independently at launch and recorded afterwards; the software does not prescribe a seed. The schema fingerprint identifies the parameter definitions in force when the run was configured, and the code entry the source revision that executed it.

Parameter Credibility (run values against the schema's documented evidence)

11 parameters, 1 changed from default; 9 within documented evidence, 0 extrapolations, 0 inadmissible, 0 numeric without a documented range, 1 switches/categories.

Parameter	Run value	Default	Unit	Credibility	Evidence for the range assessed

Chemotaxis Strength (λ_chem) (lchem)	100	500	Arbitrary Energy Units	within documented evidence	expert_judgement: Merks RMH et al. (2008); Merks et al. 2008, Methods (Cellular Potts model; chemotaxis; chemoattractant dynamics)

Credibility places each executed value against the three range tiers the schema documents: the implementation-admissible range (outside = inadmissible), the evidence-supported range (outside = extrapolation beyond the documented evidence) and the recommended range for this model's context of use. The evidence column names the basis and citation the schema records for the supported range; *not recorded* means the bound exists but its justification has not been documented, which is itself an audit finding. Numeric parameters without any documented range are listed as unassessable; switches and categories are not range-assessed. Full detail for every parameter is in parameter_audit.json inside the archive.

Per-replicate Archive Validation

Replicate	built-in SED-ML	manifest	biosimulators
angio_lchem_sweep_combo001_rep01	PASS	PASS	PASS (0 errors, 1 warning group)
angio_lchem_sweep_combo001_rep02	PASS	PASS	PASS (0 errors, 1 warning group)
angio_lchem_sweep_combo001_rep03	PASS	PASS	PASS (0 errors, 1 warning group)
angio_lchem_sweep_combo001_rep04	PASS	PASS	PASS (0 errors, 1 warning group)
angio_lchem_sweep_combo001_rep05	PASS	PASS	PASS (0 errors, 1 warning group)
angio_lchem_sweep_combo001_rep06	PASS	PASS	PASS (0 errors, 1 warning group)
angio_lchem_sweep_combo001_rep07	PASS	PASS	PASS (0 errors, 1 warning group)
angio_lchem_sweep_combo001_rep08	PASS	PASS	PASS (0 errors, 1 warning group)
angio_lchem_sweep_combo001_rep09	PASS	PASS	PASS (0 errors, 1 warning group)
angio_lchem_sweep_combo001_rep10	PASS	PASS	PASS (0 errors, 1 warning group)

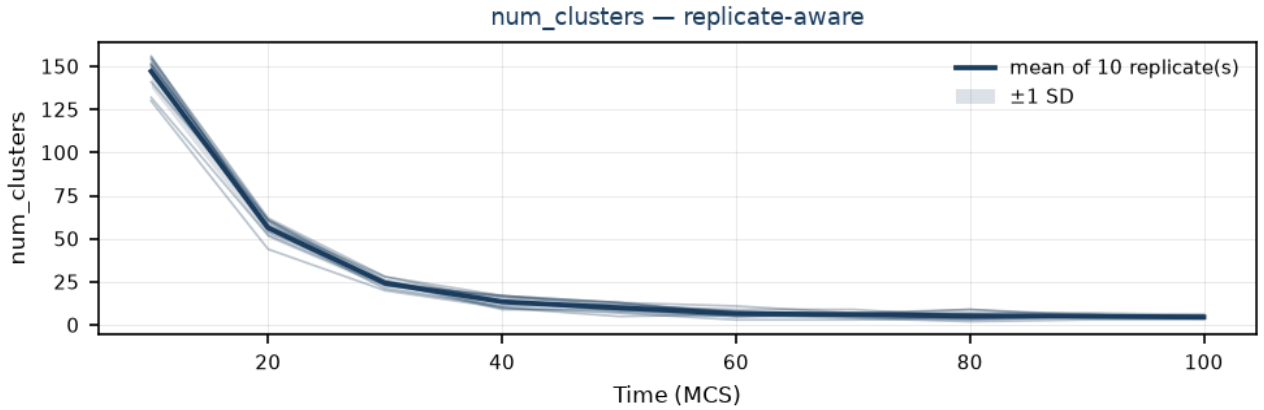
Archive Contents (per replicate)

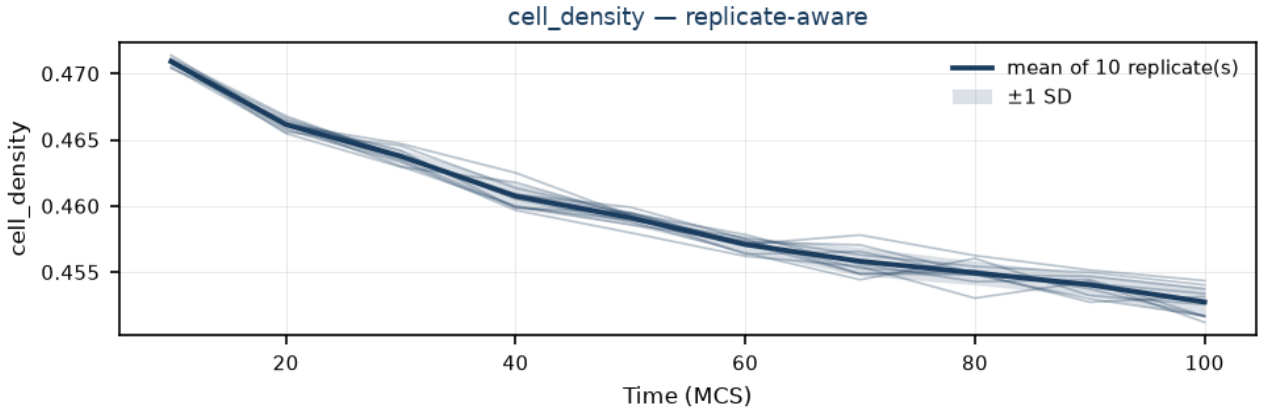
File	Size (B)	Role
logs/stderr.log	0	—
logs/stdout.log	1,846	—
manifest.xml	10,675	OMEX manifest (COMBINE archive index)
metadata.rdf	1,852	OMEX Metadata (provenance: creator, license, citation)
metrics.csv	355	Scalar signals over time (indexed by the SED-ML)
model/Simulation/angiogenesis.py	197	Bundled model file
model/Simulation/angiogenesis.xml	2,657	Bundled model file
model/Simulation/angiogenesisSteppables.py	1,260	Bundled model file
model/angiogenesis.cc3d	270	Bundled model file
model/environment.json	1,082	Bundled model file
model/model_summary.json	4,556	Bundled model artifact (model/@source resolves here)
model/parameter_audit.json	44,615	Bundled model file
model/run_model.py	789	Bundled model file
model/run_params.json	649	Bundled model file
model/src/angiogenesis_composer.py	1,116	Bundled model file
model/src/angiogenesis_process.py	99,294	Bundled model file
run_metadata.json	3,357	Run provenance / parameters
sim_params.json	649	Exact simulation parameters
simulation.sedml	5,808	SED-ML L1V4 simulation description
status.json	530	—
zarr/… (61 files)	2,058,994	Full Zarr spatial trajectory (per-timestep ndarrays)

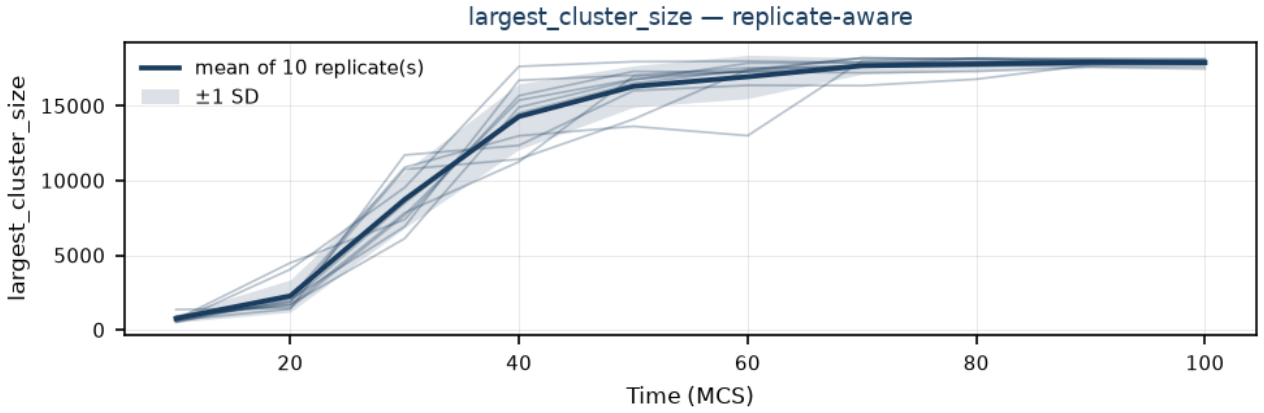
Each of the 10 replicate archives has this layout.

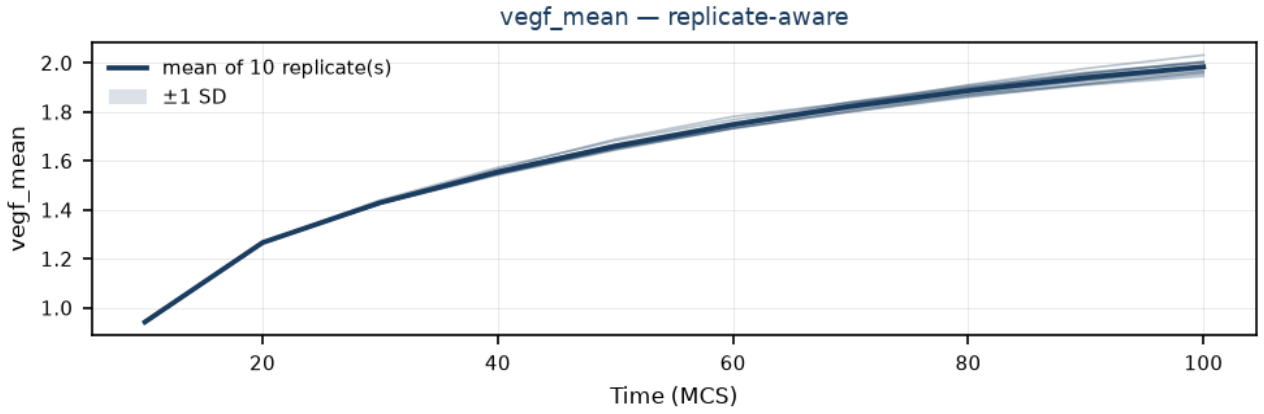
Metrics (replicate-aware, from the Zarr trajectory)

Thin = each replicate; bold = across-replicate mean; shaded = ±1 SD across replicates. Time in MCS.









Final-step spatial fields (replicate 1)

